STAT 37711 – HW 1
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Code: https://github.com/shikharsehgal1/stat37711

1.

(a) Let s be Victor's bet. At a market price of 0.60, he makes $2s/3$ if Biden wins and loses s otherwise. His expected wealth is

$$0.85(d + 2s/3) + 0.15(d - s) = d + 5s/12.$$

Since this increases with s, he bets all d.Losing everything is much worse than an equal-sized gain so linear utility does not make snse.

(b) he now maximizes

$$0.85 \log(d + 2s/3) + 0.15 \log(d - s).$$

This gives $s = 0.625d$, so he bets 62.5% of his wealth.

(c) No, not worth it.

2.

p =probability of class 1, q = predicted probability. expected cross-entropy loss is

$$R(q) = -p \log q - (1 - p) \log(1 - q).$$

$$R'(q) = \frac{q - p}{q(1 - q)}.$$

For $0 < q < 1$, the loss decreases when $q < p$ and increases when $q > p$. So it is smallest at $q = p$. Also works at $p = 0$ and $p = 1$. hence cross-entropy is a proper scoring rule.

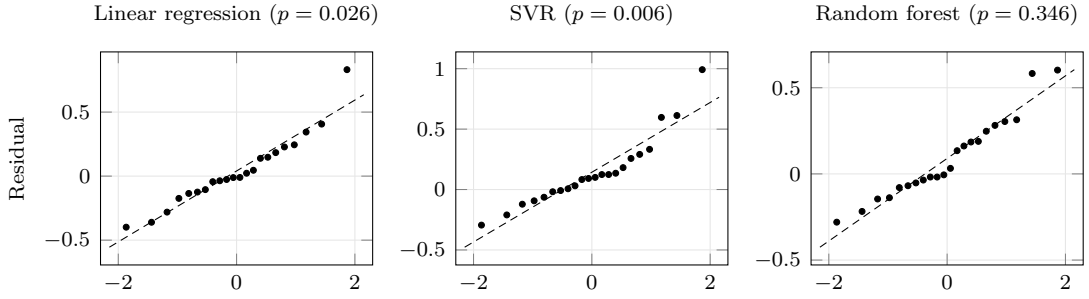
3.

Since training is for a finite number of steps exact convergence isn't necessary. A small constant learning rate may keep fluctuating around a minimum, but that is acceptable if test performance is good.

4.

Model	Test MAE	Normality p-value
Linear regression	0.1954	0.0260
SVR	0.2167	0.0055
Random forest	0.1861	0.3460

At 5%, normality is rejected for linear regression and SVR, but not random forest. This doesn't prove the random forest residuals are normal.



5.

Model	Test accuracy
sklearn logistic baseline	98.25%
PyTorch logistic	96.49%
PyTorch probit	98.25%

Regression- Part (a) uses mse (b) uses a Student-t loss with 5 dof

$$L = 3\log(1 + \text{error}^2/5)$$

Model	Test MAE
sklearn linear baseline	0.1954
PyTorch linear (squared error)	0.1954
PyTorch linear (Student-t)	0.1941

The logistic model is slightly worse than the sklearn baseline, while probit matches it.

The squared-error model matched the baseline. Student-t did slightly better, though the difference is small. Its loss is less sensitive to outliers.